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Unit “ 3 : Session “ 1 : SLO “ 1 | Stacks “ Introduction

■ I. Fill in the Blanks

- A stack is a LINEAR (LIFO) data structure.
- The two main operations of a stack are PUSH and POP.
- The operation that adds an element to a stack is called PUSH.
- The operation that removes the top element from the stack is called POP.
- The operation that returns the top element without removing it is called PEEK.
- If a stack is full and a push operation is attempted, it is called OVERFLOW.
- If a stack is empty and a pop operation is attempted, it is called UNDERFLOW.
- Stacks are usually implemented using ARRAYS or LINKED LISTS.
- In array implementation of stack, the top element is accessed using the TOP index.
- In function calls, the system uses a stack to store RETURN ADDRESSES (activation records).
- The initial value of top in an empty stack array implementation is usually -1.
- Stack is used in POSTFIX (Reverse Polish) notation to evaluate expressions without parentheses.
- A stack is also used in DEPTH-FIRST traversal of trees.
- The time complexity of push and pop operations in a stack is $O(1)$.
- Reversing a string using stack works by PUSHING each character and then POPPING them.